ROBOCUPJUNIOR RESCUE SIMULATION 2026

TEAM DESCRIPTION PAPER

Danesh Auriga

Abstract

This paper presents the design and implementation of an autonomous two-wheeled robot developed for efficient exploration, mapping, localization, navigation, and wall-token detection in simulated environments. The software is implemented in Python and integrates LiDAR, GPS, IMU, wheel-motion data, and three RGB cameras. The environment is represented as a high-resolution occupancy map and explored through A*-based path planning. A graph-based navigation strategy extends the standard grid representation by detecting traversable points between grid nodes, enabling the robot to pass through narrow passages that would otherwise remain inaccessible. For visual perception, three downward-angled cameras provide simultaneous coverage of the front, left, and right walls, improve the distinction between real and fake victims, and increase the visibility of cognitive targets at short distances. A deep-learning-based detection pipeline identifies wall tokens and estimates target centers for subsequent classification. The system was evaluated across 70 standard worlds, and the selected version achieved the best overall performance, with an average score of 1536.94 and a mean map-correctness percentage of 88.28%. These results demonstrate the effectiveness and stability of the proposed integrated approach.

1. Introduction

a. Team

Amirsam Eftekharinia (Computer Vision)

- Role and responsibility: Computer Vision, Evaluation
- Experience: Two years of robotics studies, two years of participation in the Rescue Simulation League

Ryan Feizbakhsh (Mapping)

- Role and responsibility: Mapping
- Experience: Three years of robotics studies, two years of participation in the Rescue Simulation League and one year of participation in Rescue Line League

Artin Taleei (Computer Vision)

- Role and responsibility: Computer Vision
- Experience: Two years of robotics studies, one year of participation in the Rescue Simulation League and one year of participation in Soccer Simulation League

Radin Mohammadian Asl (Navigation and Decision Making)

- Role and responsibility: Navigation and Decision Making
- Experience: Two years of robotics studies, one year of participation in the Rescue Simulation League and one year of participation in Soccer Simulation League

2. Project Planning

a. Overall Project Plan

The objective of this project is to strengthen the team members' scientific knowledge, technical skills, teamwork, and problem-solving abilities through a practical and creative experience. It also aims to introduce the team to current challenges and emerging fields in science and technology worldwide. By participating in the prestigious RoboCup competitions, the team seeks to gain valuable experience and achieve first place.

The project began with learning fundamental skills, such as the Python programming language and various libraries, including NumPy and OpenCV. After reviewing some basic mathematics, such as the two-dimensional coordinate system, and basic trigonometry and geometry, the team learned how to work with the simulator, build the robot, create the map, and become familiar with the league rules. The main development process for the robot then began. The stages of the project are listed below step by step:

Figure 1. Project Plan

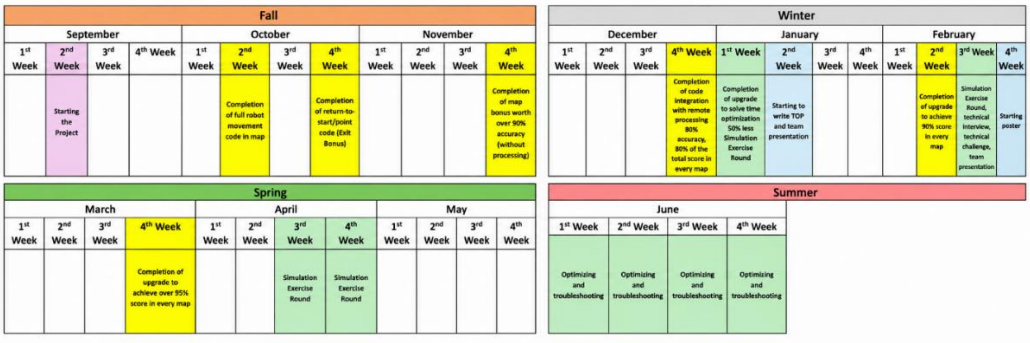


Figure 2. Deadlines and Schedules

b. Integration Plan

Our team operates in a way that, although each subsystem is primarily implemented and maintained by its designated members, all team members remain familiar with the designs and implementations of the other subsystems. Different ideas and approaches are discussed collectively, and through constructive feedback and evaluation, the most suitable solution is selected and implemented by the members responsible for that subsystem. This approach facilitates the integration of different modules and codebases, as every team member has a general understanding of the functionality of the other components.

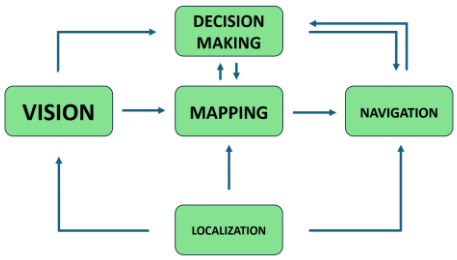


Figure 3. Robot Major Modules and their relationships

Throughout the robot software development process, special attention has been given to maintaining a modular and function-oriented code structure. This significantly reduces the time required for integration, testing, and debugging.

In addition, the use of the **Git** version control system throughout development has greatly simplified code integration, collaboration, and backup management.

3. Robot Design

The robot used in the simulation environment is equipped with two wheels, each fitted with an encoder sensor. These encoders are used to measure the distance traveled by the robot.

The robot is equipped with a LiDAR sensor for detecting obstacles, walls, and objects in the environment; a GPS sensor for localization; and an IMU sensor for determining the robot's orientation at any given time. In addition, it uses three cameras (with a resolution of 40 × 32 pixels) mounted on the left, right, and front sides of the robot for detecting tokens, identifying different colored tiles, and recognizing holes. The camera resolution was selected due to the limited sensor budget of the robot while still providing sufficient visual information for victim detection.

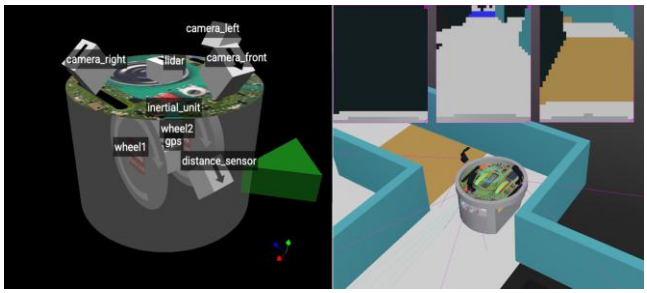


Figure 4. Robot Design

Our innovation in choosing the placement of the cameras is that we have positioned them on top of the robot at 0.65 radians angle.

The purpose of this is twofold:

1- In this configuration, fake victims appear in the camera in an unusual and different way from real victims, thus making it possible to detect whether a victim is fake using only the camera, saving time consumed by the robot.

2- In this case, when we approach cognitive targets, a larger portion of them is captured within the image, increasing the likelihood of detecting them.

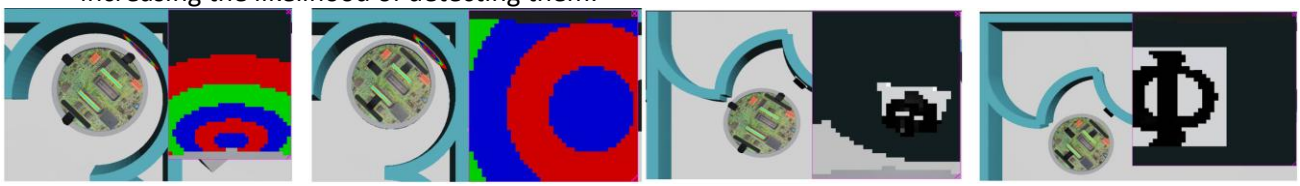


Figure 5. Difference between downward-angled and straight camera

4. Software

a. General software architecture

Several well-known and practical algorithms in the field of robotics have been used in the software, including **deep neural networks**, the **A* search algorithm**, **occupancy-grid mapping**, and **closed-loop PID control**.

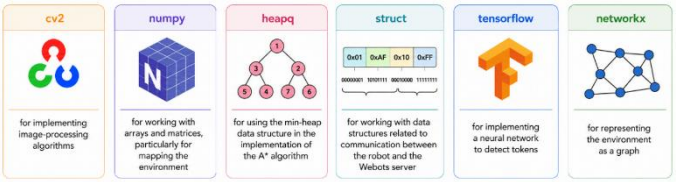


Figure 6. Libraries used in robot's software

The general operation of the robot can be described as follows. From the moment it starts, the robot divides the environment into $3\ cm\ \times\ 3\ cm$ tiles.

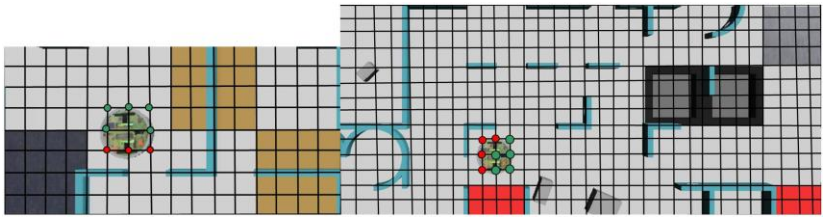


Figure 7. Dividing space into 3cm tiles

While moving, it continuously maps the environment, detects surrounding walls and obstacles, and stores this information in a two-dimensional array. The robot also records the tiles it has already visited in another two-dimensional array.

Whenever the robot reaches a tile, it examines the eight neighboring tiles. Any tile that is accessible and has not previously been visited is added to a list of future targets. At each step, the robot extracts the most recently added element from this list and selects it as its current target. It then uses the A* algorithm to calculate a path from its present position to the selected target and follows this path until it reaches the destination. The same process is then repeated. This procedure continues until the list of future targets becomes empty. At that point, there are no unexplored areas remaining on the map. The robot therefore returns to its starting point and sends an "exit" command to the server. In addition, whenever a token is detected within the robot's camera frame during the exploration process, the robot stops for one second, reports the detected symbol, and then resumes its operation.

b. Navigation

Robot Movement.

The robot's basic movements toward a target (we call it "go_to_xy") consist of a combination of in-place rotation and forward or backward motion. To maintain the robot's accuracy during different movements, a PID-controller structure has been used. However, in practice, the proportional gain (K_p) alone was enough for us.

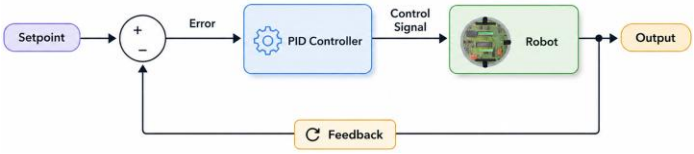


Figure 8. Feedback control structure

We now turn to the explanation of the algorithm implemented for reaching a nearby destination (local navigation). In the robot's control system, the error is defined as the difference between the robot's current orientation (θ: yaw), which is measured by the IMU sensor and the angle of the line connecting the robot's current position to the destination point (β). $\Longrightarrow error = \theta - \beta$

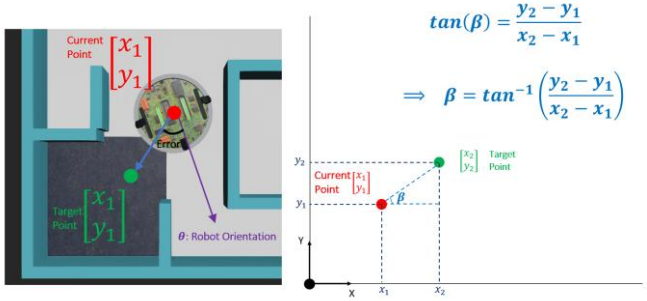


Figure 9. Calculating the orientation of the line that connects robot's current point to the destination

This error is then multiplied by an appropriate proportional gain (K_p). Based on the equations below, the control system generates the appropriate command for each wheel. In simple terms, the robot continuously attempts to adjust its orientation toward the destination within a closed-loop control structure.

$$v_r = M + K_p \times error$$
$$v_l = M - K_p \times error$$

(M: maximum velocity of the robot, v_r: right wheel speed, v_l: left wheel speed)

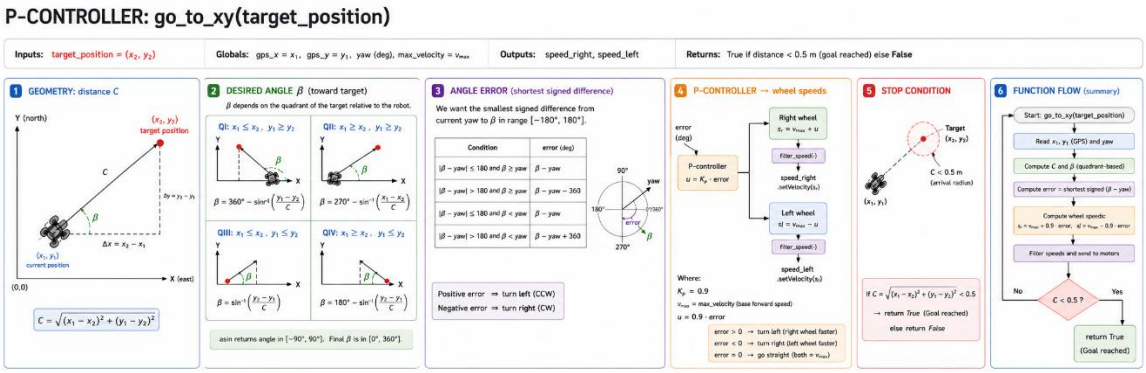


Figure 10. Complete robot movement pipeline

In the design of our controller, tuning the proportional gain (K_p) is an important task, as an inappropriate value can reduce the quality of the system's performance. The effect of changing (K_p) on the system performance, error, and overshoot is shown in the figure below.

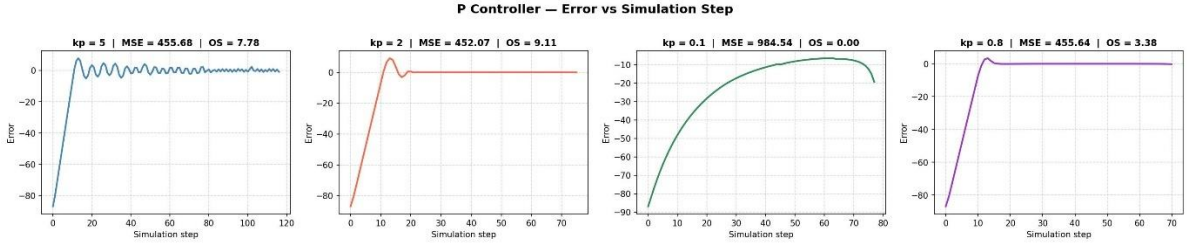


Figure 11. As you can see the gain of 0.8 (which is the value used in our controller) has the best Mean Square Error (MSE) and Overshoot (OS) with respect to the others

Path Planning.

Our robot uses the A* search algorithm to find a path from an arbitrary point **A** to an arbitrary point **B**.

The **heapq** library is used in our A* implementation. This library allows us to efficiently add tile coordinates to our lists or remove them when needed. Using this library instead of manually searching through the open list improves the execution speed of the code. More specifically, the library stores the data in the Open Set using a min-heap data structure. As a result, the time complexity of the relevant operations is reduced from $O(n)$ to $O(\log n)$.

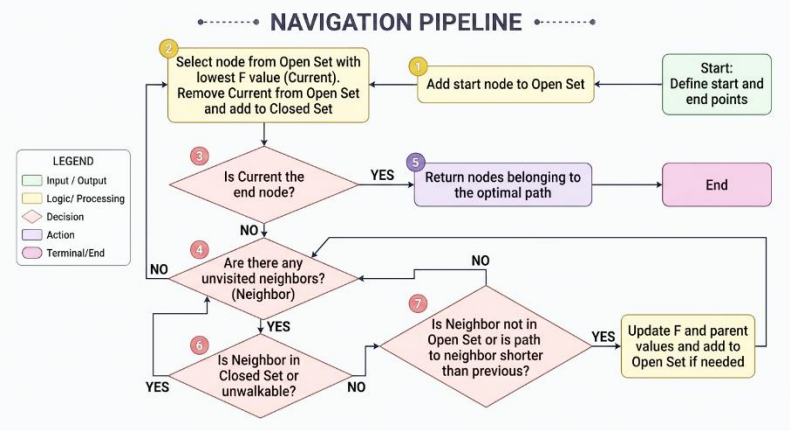


Figure 12. A algorithm flowchart*

Narrow-Passage Navigation Algorithm.

During the testing and development of our code, we encountered situations in which the robot may need to pass through narrow corridors or pathways that cannot be traversed solely by moving along grid points. An example of such a situation is shown in the figure below.

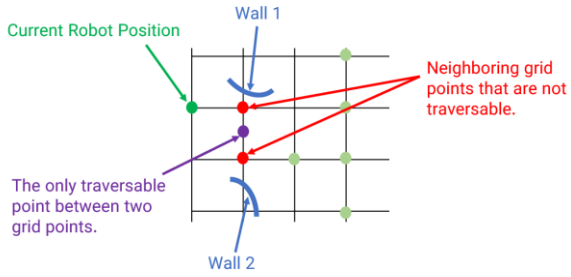


Figure 13. Visualizing narrow-passage scenario

As shown in the figure, the only path between the two walls passes through the purple point, which is located **between** two standard grid points. This limitation led us to move beyond the grid-based representation and model the environment as a graph instead. We refer to this approach as "**graph-based navigation**". If two adjacent grid points are not traversable, the points between them are examined using a step size of 0.1 cm. If a traversable point is found, it is added to the graph as a new vertex, and the robot's current vertex is defined as one of its neighbors. This process is repeated whenever the robot reaches a new graph vertex.

In this way, the map of the environment is constructed as a graph while the robot is moving. Each edge in the graph represents the robot's ability to move between the two corresponding vertices. This graph-based map is then used for path planning (A* algorithm).

The desired graph is implemented using the **NetworkX** library.

By constructing a graph-based representation of the environment, the robot is able to navigate through narrow passages. The figure below shows an example of a narrow passage that would not have been traversable without this algorithm. The red point represents the position outside the standard grid.

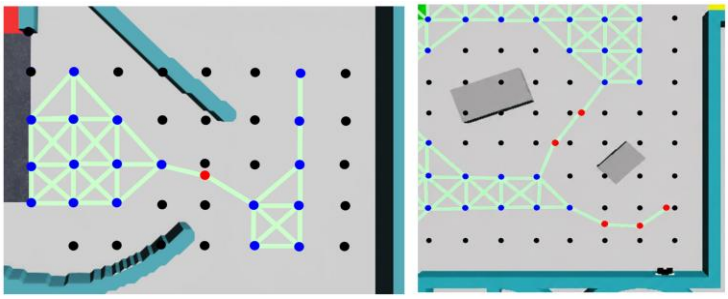


Figure 14. Occurrence of narrow-passages in real maps and how our algorithm with off-grid nodes can solve the problem

c. Wall Token detection

The system must operate reliably under varying viewpoints, rotations, and perspective distortions while maintaining real-time performance during exploration. To achieve this, our team developed a deep-learning-based vision pipeline using YOLO26-Pose. The model detects wall tokens and estimates the center point of cognitive targets.

Camera Configuration.

During early testing, a forward-facing camera configuration showed two main limitations. First, real victims and fake victims often appeared similar. Second, when the robot was close to a wall, the outer rings of cognitive targets were often outside the vertical range of the image.

To solve these problems, the cameras were tilted downward. This configuration exposes the side surfaces of fake victims. In addition, the downward viewing angle increases the visible wall area at close distances. The angled camera position introduces perspective distortion in the vertical direction. However, this effect is compensated **through image warping techniques** that map image coordinates to real-world proportions.

The final camera orientation was selected after evaluating multiple configurations and comparing victim recognition performance in representative competition scenarios.

A custom dataset was collected directly from the simulation environment. Images were recorded while the robot navigated through different worlds and encountered various wall tokens and environmental features.

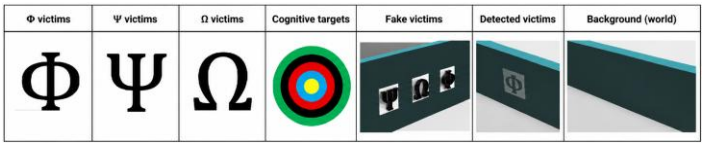


Figure 15. Dataset contains the following cases

The "Background" class contains walls and other environmental elements that do not correspond to valid wall tokens, helping the model distinguish targets from non-target objects.

Special attention was given to collecting samples under a wide range of conditions, including different viewing angles, large victim rotations, different distance from the wall, partial visibility and occlusions, and various positions within the camera frame

The dataset was collected in a way to cover the full rotational range. Additional samples were recorded from different locations and orientations to improve the model's ability to generalize to unseen environments. The final dataset was manually labeled using bounding boxes for all classes. For cognitive targets, an additional center keypoint was annotated to support precise center localization.

Synthetic Cognitive Target Generation.

The total number of possible color combinations for cognitive targets is $5^5 = 3125$ and hence, Collecting and manually labeling all combinations would be time-consuming and inefficient. Therefore, a synthetic data generation pipeline was developed.

The generator automatically creates cognitive target images with editing the labeled images and creating random color rings. This approach significantly increases dataset diversity while reducing manual labeling

effort.

Annotation Strategy.

The victim detection model is trained using the pose-estimation variant of YOLO26. The network is configured to detect four classes: Φ victims, Ψ victims, Ω victims, Cognitive targets

For letter victims, standard bounding-box annotations are used. Since only the victim type and location are required, no keypoints are assigned to these classes. For cognitive targets, a single keypoint is additionally annotated at the center of the target.

After detection, the center position is used to extract color information from the concentric rings. The colors are then converted into their corresponding numerical values and summed to determine the cognitive target type. After training, the network is exported from Ultralytics format to ONNX and simplified then converted into a TensorRT engine.

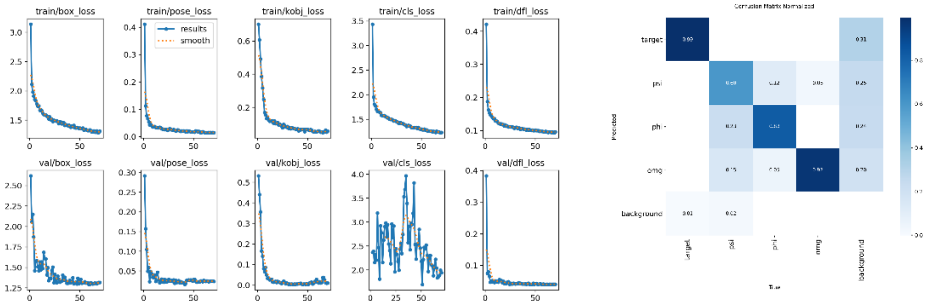


Figure 16. Training results and Confusion Matrix

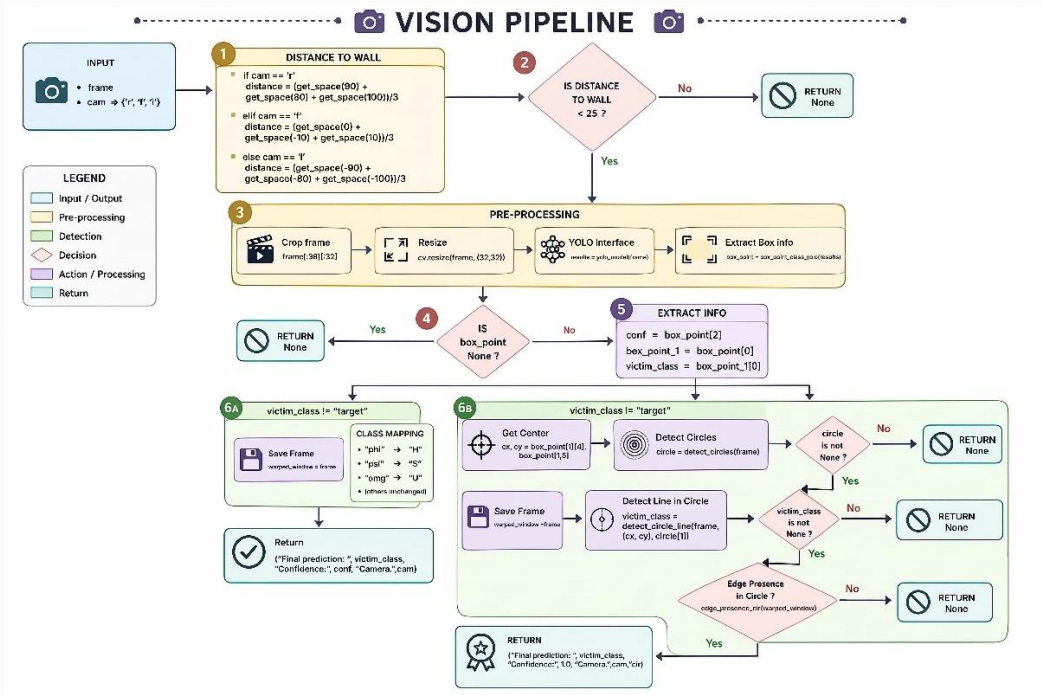


Figure 17. Computer Vision algorithm pipeline

TensorRT optimization significantly reduces inference latency and increases frame rate, allowing victim detection to run continuously during navigation without affecting other subsystems.

Figure 18. Victim detection (Φ, Ψ, Ω)

d. Mapping

This section examines how the robot determines its position within the environment, as well as the shape and characteristics of its surroundings.

Localization.

We designed a localization system based on the robot's equations of motion. In this system, the changes in the robot's position along the x-axis and y-axis can be calculated based on the rotational speeds of its wheels and its orientation.

Therefore, by adding Δx and Δy to the previously estimated position values, the robot can obtain an estimate of its new position. The equations of motion for our two-wheeled robot are presented below.

$$\Delta x = \left(\frac{v_r + v_l}{2}\right) \times \Delta t \times \cos(\theta)$$

$$\Delta y = \left(\frac{v_r + v_l}{2}\right) \times \Delta t \times \sin(\theta)$$

(Δx: Displacement along the x-axis, Δy: Displacement along the y-axis, L: Distance between two wheels, Δt: Time difference, v_r: Right wheel velocity, v_l: Left wheel velocity, θ: IMU yaw measurement)

For the equations, since the wheel velocities are commanded directly (also can be measured by encoders) and the time interval and robot orientation are known, the change in the robot's position can be calculated at each step.

Mapping.

The LiDAR sensor provides the distances to obstacles and walls at different angles around the robot, covering a full 360-degree field of view. Next, trigonometric transformations based on the sine and cosine relations are used to determine the positions of obstacle points in the robot coordinate system, also referred to as the "local coordinate system". However, this alone is not sufficient to construct an accurate map of the environment, since the map must be represented relative to a **global coordinate system**.

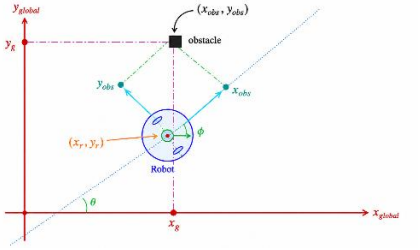


Figure 19. Calculating obstacle coordinates in the global frame

The exact equations used to convert the sensor data into the obstacle coordinates in the global coordinate system are presented below.

$$\begin{bmatrix} x_{obs_{local}} \\ y_{obs_{local}} \end{bmatrix} = \begin{bmatrix} L \times \cos(\alpha) \\ L \times \sin(\alpha) \end{bmatrix} \rightarrow \begin{bmatrix} x_{obs_{global}} \\ y_{obs_{global}} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x_{obs_{local}} \\ y_{obs_{local}} \end{bmatrix} + \begin{bmatrix} x_{robot} \\ y_{robot} \end{bmatrix}$$

Here, α denotes the angle of the corresponding LiDAR laser beam, and (L) represents the distance measurement returned by the LiDAR sensor at angle α. The remaining variables are defined in the above figure.

Finally, the generated map is stored in a two-dimensional array. The resolution of our map is set to **1 mm**. Two examples of the outputs generated by the environment-mapping system are shown below.

Figure 20. Some results of the mapping algorithm after robot explores the environment

During navigation, the robot maps the environment and records colored tiles and each tile with its associated room. Before exiting, it converts the map into 12 cm × 12 cm tiles, generates a 5 × 5 matrix for each tile, and combines them into the final array sent to the server.

Figure 21. construction of server-type map from our 2d-array

5. Performance evaluation

The robot's performance is evaluated through comprehensive tests designed to closely resemble the conditions of the actual competition day across different arenas. Certain parameter settings may significantly improve performance in a specific arena while reducing the robot's overall performance across a wider range of environments. To avoid this issue, each version of the code is tested on several different maps, and the average of the resulting scores is used as the basis for decision-making.

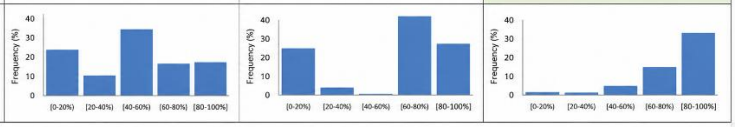
	Version 1	Version 2	Version 3
Mean of total score	1082.82	1238.81	1536.94
Mean of Map Correctness Percentage	59.79%	55.73%	88.28%
Mean of Exit Bonus	60.35	43.98	73.70
Total Score Variance	827.99	906.25	1123.59
Map Correctness Percentage Variance	0.37	0.4337	0.1688
Map Correctness Percentage Distribution			

Figure 22. Evaluation results

As the data indicate, Version 3 has a higher average score, a higher **Map Bonus** percentage, and a higher exit score. In addition, according to the histogram, Version 3 does not produce a large number of very low-scoring runs. Therefore, we conclude that Version 3 is more stable and is the preferred version.

6. Conclusion

In this project, we aimed to design and implement a capable robot for exploration, mapping, and decision-making in a simulated environment by drawing on the technical expertise of the team members and employing intelligent methods. The results demonstrate that the careful integration of sensors, learning-based methods, and motion-control techniques can improve the robot's accuracy, speed, and stability. Furthermore, evaluations conducted across different maps indicate that the system has a suitable level of reliability and can be further optimized in the future. Continuing this work can enhance the team members' technical expertise and contribute to improved results in future competitions.

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